

AFRISOLVE HUB

Project Plan & Technical Specification

Connecting Problem Posters with Skilled Developers across Ghana

1. Executive Summary

AfriSolve Hub is a web-based marketplace that connects Ghanaians with everyday problems to skilled developers who can build software solutions. Problem posters (individuals, businesses, or researchers) submit challenges; approved developers (individuals or teams) pick them up, build a solution, and engage through a fair, transparent financial commitment model. The platform manages the entire lifecycle: problem submission, team formation, approval, progress tracking (0%–100%), and eventual handover.

To ensure genuine commitment from all parties, both the problem poster and the assigned developer independently assess the total project cost, then each contributes 20% of that assessed cost as a down payment held in a third-party escrow account managed by the platform. If either party withdraws from the project, the down payment funds are transferred to the remaining party as compensation. Payments are processed securely through Paystack, supporting mobile money and card payments.

The platform is built entirely on Python Django, using Django's templating engine for the frontend. User identity is verified via Ghana Card during registration to ensure accountability and reduce fraudulent activity.

2. Problem Statement

In Ghana, many communities and small businesses face persistent issues that could be solved with simple software — yet they lack access to developers. Simultaneously, many young developers need real-world projects to build their portfolios and earn income. There is no structured marketplace linking these two groups with a fair, accountable financial model that protects both sides. Additionally, the absence of identity verification on digital platforms has led to high rates of abandonment and dishonesty in collaborative projects.

3. Solution Overview

A full-stack Django web application (backend + Django templating frontend) that:

- Allows verified, Ghana Card-registered users to post problems.
- Displays a public problem board with tags showing poster, status, and assigned developer/team.
- Lets developers (individuals or teams) request approval to work on a problem.
- Locks a problem once a developer/team is assigned.

- Tracks progress in five stages: 0% (Accepted), 30%, 50%, 80%, 100% (Completed).
- Implements a down payment escrow model (20% each from poster and developer) managed via Pystack.
- Hosts a real-time chat room per problem for the developer, poster, and admin.
- Provides dashboards for posters, developers, and admins.

4. Core Features

4.1 User Management

- Registration/Login (email/password, optional social authentication)
- Roles: Problem Poster, Developer, Admin
- Profile pages with reputation score, past projects, and ratings
- Ghana Card Verification (mandatory during registration — see Section 4.1.1 below)

4.1.1 Ghana Card Verification

To ensure accountability and reduce fake accounts, every user must verify their Ghana National Identification Card (Ghana Card) during registration. The process is:

- User fills in their Ghana Card number in the format GHA-XXXXXXXX-X during sign-up.
- User uploads a clear photo of the front of their Ghana Card.
- The system stores the Ghana Card number and image securely (encrypted at rest).
- An admin or automated verification service reviews and approves the submission.
- Until the Ghana Card is verified, the user account is in a ‘Pending Verification’ state and cannot post problems or apply for projects.
- Upon successful verification, the account is activated and the user receives a ‘Verified’ badge on their profile.

Note: Ghana Card verification prevents duplicate accounts, builds trust in the marketplace, and provides a legal link between platform activity and a real national identity. All ID data is stored encrypted and access is restricted to admins only.

4.2 Problem Lifecycle

- Post a Problem: Title, category, detailed description, location/area, potential business impact, attachments.
- Public Board: Filterable list of ‘open’ problems (only those not yet assigned).
- Locked/Assigned Problems: Visible to the assigned developer/team and the poster.
- Progress Updates: Developer updates progress percentage with notes; poster can view in real time.

4.3 Developer Request & Approval

- Apply: Individual developer or Team Leader applies for a specific open problem.

- Team Management:
 - Create a team (name, members).
 - Team Leader invites other developers by email/username.
 - Team applies as a unit; if approved, all members are tagged.
- Admin Review: Admin approves or rejects the request.
- Once approved, problem status changes to ‘Assigned/Locked’ and the start date is recorded.
- Developer/Team confirms estimated duration and acknowledges they will communicate with the poster.

4.4 Progress Tracking

Each assigned problem has a dashboard with a progress selector offering five milestones:

- 0% – Work Started / Accepted
- 30% – Initial Development
- 50% – Mid-point
- 80% – Near Completion
- 100% – Completed & Delivered

Each update requires a short note and an optional file upload. The problem poster receives an in-app notification and email alert for every milestone update.

4.5 Project Cost Assessment & Down Payment Escrow Model

This section replaces the previous 60/40 profit-sharing model. The new model is designed to ensure genuine financial commitment from both parties before work begins, protecting both the developer and the problem poster.

4.5.1 How It Works

- Cost Assessment: After a developer/team is approved for a problem, both the developer and the problem poster independently assess or mutually agree on the total estimated cost of building the solution.
- Down Payment Calculation: Each party (developer and problem poster) is required to pay 20% of the agreed project cost as a down payment.
- Escrow Deposit via Paystack: Both payments (20% from the developer + 20% from the poster) are processed through Paystack and held in a third-party escrow account controlled by the platform administrator.
- Project Work Begins: Only after both down payments are confirmed does the project status change to ‘In Progress’ and the developer officially begins work.
- Completion & Release: Upon successful delivery and sign-off, the escrowed funds are released back to their respective owners, or applied toward the final agreed payment.

4.5.2 Quit/Abandonment Protection

If the Developer quits or abandons the project:

→ The developer's 20% down payment is forfeited and transferred to the problem poster.

If the Problem Poster quits, cancels, or refuses to cooperate:

→ The poster's 20% down payment is forfeited and transferred to the developer.

This creates a financial stake that discourages both parties from walking away.

4.5.3 Platform & Agreement

- The platform acts as a neutral third party / escrow manager.
- Both parties digitally sign a Project Agreement before the down payment is processed.
- The agreement outlines: total project cost, timeline, deliverables, and the down payment forfeiture terms.
- Platform can charge a small escrow management fee (e.g., 2–5% of the total project cost) for sustainability.
- In case of disputes, the admin/arbitrator reviews the project evidence (chat logs, progress updates, deliverables) and decides the outcome.

4.6 Chat System (Per-Problem Chat Room)

Each problem on the platform has a dedicated chat room that enables real-time, threaded communication between the three key stakeholders: the Problem Poster, the assigned Developer (or team members), and the Platform Administrator.

4.6.1 Chat Features

- Real-time messaging powered by Django Channels and WebSockets.
- Participants: Problem Poster, Developer/Team Members, and Admin(s) on that specific problem.
- Message history stored in the database and viewable at any time by participants.
- File attachments in chat (images, documents, code snippets).
- Admin moderation: Admin can view all chat rooms for oversight and can intervene in disputes.
- Read receipts so users know when their messages have been seen.
- Notifications: Users receive in-app and email notifications when they receive a new message.

4.6.2 Chat Privacy

- Chat rooms are private to the assigned developer, the poster, and admin — not visible to the public.
- Chat logs are retained for dispute resolution purposes.
- Admin can view logs if a formal dispute is raised.

4.7 Payment Integration (Paystack)

Paystack is the primary payment processor for AfriSolve Hub. It handles all monetary transactions on the platform, including down payments, platform fees, and any future payout flows.

4.7.1 Payment Flows

- **Down Payment Collection:** When both parties confirm project cost, the system generates two Paystack payment links (one for the developer, one for the poster), each for 20% of the agreed cost.
- **Platform Fee:** A platform usage or escrow management fee can be charged at the time of down payment or project completion.
- **Escrow Management:** Funds are collected via Paystack and held in the platform's designated account until project completion or forfeiture conditions are triggered.
- **Payouts:** When funds are to be released or forfeited, the admin processes the transaction through Paystack's Transfer API.

4.7.2 Supported Payment Methods (Ghana)

- **Mobile Money:** MTN Mobile Money, Vodafone Cash, AirtelTigo Money (all supported by Paystack).
- **Bank Cards:** Visa and Mastercard debit/credit cards.
- **USSD:** For users without smartphones or internet access.

4.7.3 Paystack Integration Points

- **Paystack Inline (pop-up checkout)** for in-browser payment initiation.
- **Paystack Webhooks** for real-time payment event notifications (payment confirmed, failed, refunded).
- **Paystack Transfer API** for sending funds from the platform account to users.
- **Paystack Split Payments** optionally, to split platform fees automatically at the point of collection.

4.8 Notifications & Communication

- **In-app messaging** via the dedicated per-problem chat room (Section 4.6).
- **Email alerts** for status changes, approvals, new problems, new messages, payment confirmations.
- **In-app notification bell** for real-time alerts within the platform.
- **Optional SMS** via Africa's Talking (for Ghana numbers) for critical alerts (e.g., payment received, project approved).

4.9 Admin Dashboard

- **Manage users:** view, suspend, verify Ghana Cards, assign roles.
- **Manage problems:** review, approve, reject, close.

- Manage applications: approve or reject developer/team requests.
- Escrow management: view down payment statuses, process releases and forfeitures.
- Chat oversight: view chat rooms for any problem, especially during disputes.
- Analytics & reports: problems by region, completion rate, active developers, revenue.

5. User Workflows

5.1 Problem Poster

Sign up → submit Ghana Card for verification → await activation. Complete profile. Click “Post a Problem” → fill form → submit. Receive notification when a developer/team applies. Jointly assess project cost with the assigned developer. Pay 20% down payment via Paystack → wait for developer to do the same. Communicate with developer via the problem’s chat room. Monitor progress on the problem page. Sign off on completion → down payment funds released or applied to final settlement.

5.2 Developer (Individual)

Sign up → submit Ghana Card for verification → await activation. Complete profile (skills, portfolio). Browse “Open Problems” board. Click “Request to Solve” → enter estimated duration, confirm commitment. Wait for admin approval → problem locked to this developer. Jointly assess project cost with the poster. Pay 20% down payment via Paystack. Update progress at each milestone (0%–100%) with notes. Deliver solution, sign off → down payment released.

5.3 Team Leader

Sign up → verify Ghana Card → Create Team → invite members. Browse problems; apply as a team. Admin approves → entire team tagged to the problem. Team leader assesses project cost with the poster and pays team’s 20% down payment. Team leader updates progress; all members can view and chat. Team leader signs off on completion.

6. Tech Stack

AfriSolve Hub is built entirely on a Python Django stack. Django’s built-in templating engine powers the frontend, eliminating the need for a separate JavaScript framework. This simplifies deployment, reduces complexity, and keeps the entire codebase in one unified project.

Category	Technology / Details
Backend	Python 3.11+, Django 4.2+, Django REST Framework (for internal API endpoints and Paystack webhooks)
Frontend	Django Templating Engine, HTML5, CSS3, Bootstrap 5 (responsive UI), minimal vanilla JavaScript (AJAX for partial page updates)

Database	PostgreSQL (primary relational database), Redis (caching, session storage, task queue)
Real-time Chat	Django Channels + WebSockets (for per-problem chat rooms and live progress notifications)
Task Queue	Celery + Redis (email notifications, background jobs, payment status polling)
File Storage	Local filesystem (development) / AWS S3 or DigitalOcean Spaces (production)
Authentication	Django Allauth (social auth optional) + Django session-based auth (no JWT needed for template-based frontend)
Payments	Paystack (down payment collection, escrow management, mobile money, card payments, Transfer API)
SMS (optional)	Africa's Talking API (Ghana numbers, critical notifications)
ID Verification	Ghana Card upload + admin/manual review (with future integration to NIA API)
PDF / Agreements	WeasyPrint or ReportLab (generate project agreement PDFs for digital signing)
Hosting – App	Render / DigitalOcean App Platform / AWS EC2 (single Django deployment)
Hosting – DB	Managed PostgreSQL (DigitalOcean, Supabase, or AWS RDS)
Domain	.com.gh domain for local Ghanaian trust

7. Three-Month Development Breakdown

Month 1: Foundation & Core Functionality

- Week 1–2: Set up Django project, custom User model with roles, session-based authentication, Ghana Card upload field, registration and login pages using Django templates, basic responsive layout with Bootstrap 5.
- Week 3: Ghana Card verification admin workflow; Problem model & views (create, list open problems); public problem board template with filters.
- Week 4: Developer application model & logic; ‘Request to Solve’ view; admin approval workflow; problem locking mechanism.

Month 2: Teams, Chat, Payments & Progress Tracking

- Week 5: Team model (creator, members, team name); invite/accept flow via email; team application to problem.

- Week 6: Paystack integration — down payment collection flow, webhook handling, escrow status tracking; Project Cost Assessment form; digital project agreement template and PDF generation.
- Week 7: Django Channels setup; per-problem chat room; real-time WebSocket messaging; chat history storage; read receipts and notifications.
- Week 8: Progress tracking model (0–100% milestones); progress update views; dashboards for poster and developer (Django template-driven); notification system (email + in-app).

Month 3: Polish, Testing & Deployment

- Week 9: Admin dashboard (stats, user management, Ghana Card approval queue, escrow management, chat oversight); moderator features.
- Week 10: Security audit — CSRF protection, rate limiting, input validation, role-based access control; Paystack Transfer API for forfeitures and releases; data encryption for Ghana Card images.
- Week 11: End-to-end testing, bug fixes, responsive design review, performance optimisation (query optimisation, caching with Redis).
- Week 12: Deployment to production, user acceptance testing (UAT) with real Ghanaian testers, documentation, beta launch.

8. Suggestions for Success

8.1 Legal & Governance

- Work with a Ghanaian lawyer to draft a standard Project Agreement. Clearly define what happens to the down payment in different exit scenarios.
- Require Ghana Card KYC during registration to reduce fake accounts — already built into the platform.
- Register the platform with the Ghana Data Protection Commission due to storage of national ID data.

8.2 Community Building

- Run hackathons in Accra, Kumasi, and Tamale to seed problems and developers.
- Feature “Projects of the Week” and success stories to attract both sides.
- Partner with local universities (Legon, KNUST) for developer talent.

8.3 Monetisation

- Charge a platform/escrow management fee (2–5% of project cost) on each down payment transaction.
- Offer premium developer profiles or featured problem slots.
- Charge for verified problem posts to filter serious submissions.

8.4 Trust & Safety

- Ghana Card verification is the first line of defence against fake accounts.
- The down payment escrow model creates financial accountability for both parties.
- Introduce a rating/review system after project completion.
- Provide a formal mediation process for disputes, with chat logs as evidence.

8.5 Technical Enhancements

- Integrate with GitHub/GitLab for automatic progress tracking via commit webhooks.
- Build a Progressive Web App (PWA) version for offline access and mobile-like experience.
- Build a mobile app (Flutter or React Native) later to increase accessibility.
- Integrate with the Ghana NIA API for automated Ghana Card verification in a future version.

8.6 Localisation

- Allow users to post problems in local languages (Twi, Ga, Ewe) with translation support.
- Host on a .com.gh domain for local trust.
- Display amounts in Ghana Cedis (GHS) throughout the platform.

9. Conclusion

AfriSolve Hub is a unique blend of crowdsourcing innovation and impact investing, directly tackling unemployment and everyday challenges in Ghana. By replacing the previous profit-sharing model with a transparent down payment escrow system powered by Paystack, the platform creates genuine financial accountability for both problem posters and developers — making it more than just a freelancing platform, but a trust-based startup incubator at the community level.

The addition of Ghana Card verification at registration ensures a verified, accountable user base. The per-problem real-time chat system ensures smooth collaboration between all stakeholders. Building entirely on Django (backend + Django templating for the frontend) keeps the stack unified, maintainable, and efficient for a small team to develop and deploy within three months.

With strong execution and community engagement, AfriSolve Hub has the potential to become the go-to platform for solving Ghana's problems through local software talent.